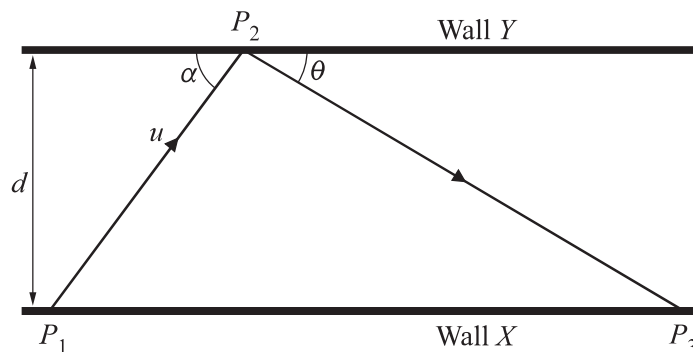




X and Y are two fixed smooth vertical walls on a smooth horizontal surface. The walls are parallel and at a distance d apart. The points P_1 , P_2 and P_3 all lie on the surface.

A particle Q is projected horizontally from the point P_1 on Wall X with speed u , and moves along the surface. The particle Q strikes Wall Y at the point P_2 . Immediately before the collision, the direction of motion of Q makes an angle α with Wall Y , where $\sin \alpha = \frac{4}{5}$. Immediately after the collision, the direction of motion of Q makes an angle θ with Wall Y . The particle Q then strikes Wall X at the point P_3 (see diagram).

The time that it takes Q to travel the distance P_1P_2 is T . The time that it takes Q to travel the distance P_2P_3 is kT .

Find, in terms of k , the coefficient of restitution between Q and Wall Y . [6]

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